



VNUHCM-UNIVERSITY OF SCIENCE
FINAL EXAMINATION
Semester 3 – Academic year 2024-2025

ARCHIVE CODE
(written by ET&QA Office)

Course name:	Calculus III	Course code:	MTH253
Time:	90 minutes	Date:	
Note: Students are [☐ allowed / ☒ not allowed] to use materials during the examination			

Full name of Student: Student ID: No:

Question 1 (2 points).

Evaluate $\iint_D \cos(\sqrt{x^2 + y^2}) dA$ where D is the top half of the disk with center the origin and radius 2.

Question 2 (2 points).

Let C be the arc of the curve given by $\mathbf{r}(t) = (t, \sqrt{3} \sin t, \sqrt{3} \cos t)$ where $0 \leq t \leq \frac{\pi}{6}$.
Evaluate $\int_C yz \sin x ds$.

Question 3 (2 points).

Let $\mathbf{F} = (x + y, xy)$ and C be positively oriented curve given by the boundary of the region enclosed by the parabola $y = x^2$ and the line $y = 2x$.

- Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$.
- Check your result using Green's theorem.

Question 4 (2 points).

Evaluate $\iint_S (x^2 z + y^2 z) dS$, where S is the hemisphere $x^2 + y^2 + z^2 = 4, z \geq 0$.

Question 5 (2 points).

Evaluate $\iint_S -x dy dz - y dx dz + z^3 dx dy$, where S is the part of the cone $z = \sqrt{x^2 + y^2}$ between the plane $z = 1$ and $z = 3$ with downward orientation.